

Adjoint regularity on the Heisenberg group

Entropy and higher integrability for measurable diffusion matrices

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Abstract

We study nonnegative adjoint solutions for arbitrary Borel uniformly positive horizontal diffusion matrices on the Heisenberg group. We prove absolute continuity and a local space-time $L \log L$ estimate, uniformly under addition of a coordinate Laplacian with coefficient in $[0, 1]$, without initial density or entropy assumptions. The proof uses an exact central regularization of scalar, vector, and tensor fields, and the bracket estimate that controls a half derivative in the central direction by horizontal energy. In the zero-viscosity case, homogeneity and a quantitative nonconcentration argument improve entropy to a positive power, both for stationary solutions and for parabolic solutions over the full time interval. An exact stationary family specifies a limitation on the admissible power exponents.

1 Setting and the entropy theorem

Fix $d \geq 1$. On $\mathbb{H}^d = \mathbb{R}^{2d+1}$ use coordinates $g = (x, y, z)$, $x, y \in \mathbb{R}^d$, and group law

$$(x, y, z)(x', y', z') = (x + x', y + y', z + z' + \tfrac{1}{2}(x \cdot y' - y \cdot x')).$$

The horizontal fields and the central field are

$$X_j = \partial_{x_j} - \tfrac{1}{2}y_j\partial_z, \quad Y_j = \partial_{y_j} + \tfrac{1}{2}x_j\partial_z, \quad Z = \partial_z.$$

Write $V = (X_1, \dots, X_d, Y_1, \dots, Y_d)$ and $m = 2d$. Then $[X_j, Y_k] = \delta_{jk}Z$, the other horizontal brackets vanish, and all these fields are divergence free for Haar measure $dg = dx dy dz$. The dilations are $\delta_r(x, y, z) = (rx, ry, r^2z)$, and the homogeneous dimension is $\mathfrak{q} = 2d + 2$. For a Borel symmetric $m \times m$ matrix $A = (a_{ij})$, set

$$L_A \varphi = \tfrac{1}{2} \sum_{i,j=1}^m a_{ij} (V_i V_j + V_j V_i) \varphi.$$

There is no additional first-order term. The coordinate Laplacian below is $\Delta = \sum_j (\partial_{x_j}^2 + \partial_{y_j}^2) + \partial_z^2$.

Theorem 1.1 (Parabolic measures, uniform coordinate viscosity). *Let $\Omega \subset \mathbb{H}^d$ be open, $T > 0$, and $0 < \lambda \leq \Lambda < \infty$. Let $(\nu_t)_{0 < t < T}$ be nonnegative finite Borel measures on Ω , weakly measurable in the sense that $t \mapsto \int \psi d\nu_t$ is measurable for every $\psi \in C_c(\Omega)$, and suppose $\nu_t(\Omega) \leq 1$ for almost every t . Set $d\nu(t, g) = dt d\nu_t(g)$. Assume A is Borel symmetric and $\lambda I_m \leq A(t, g) \leq \Lambda I_m$ for ν -almost every (t, g) . For $\varepsilon \in [0, 1]$ and a finite nonnegative Borel measure μ , suppose*

$$\int_{(0,T) \times \Omega} (\partial_t \varphi + L_A \varphi + \varepsilon \Delta \varphi) d\nu = \int_{(0,T) \times \Omega} \varphi d\mu \quad (\varphi \in C_c^\infty((0, T) \times \Omega)). \quad (1.1)$$

Then $\nu = p dt dg$ with $p \geq 0$, and for every $K \Subset \Omega$,

$$\int_0^T \int_{\Omega} p dg dt \leq T, \quad \int_0^T \int_K p \log(1+p) dg dt \leq C(T, K, \Omega, d, \lambda, \Lambda).$$

The constant is independent of A, ε, ν, μ . No initial density, initial entropy, coefficient continuity, or smallness of the ellipticity ratio is assumed.

Corollary 1.2 (Stationary measures). *Let σ be a finite nonnegative Borel measure on Ω with mass at most one. Let $A = A(g)$ be Borel symmetric with $\lambda I_m \leq A \leq \Lambda I_m$ σ -almost everywhere. If $\varepsilon \in [0, 1]$, μ_0 is a finite nonnegative Borel measure, and*

$$\int_{\Omega} (L_A \varphi + \varepsilon \Delta \varphi) d\sigma = \int_{\Omega} \varphi d\mu_0 \quad (\varphi \in C_c^\infty(\Omega)),$$

then $\sigma = p dg$ and $\int_K p \log(1+p) \leq C(K, \Omega, d, \lambda, \Lambda)$.

Proof. Apply Theorem 1.1 on a time interval of length one with $\nu_t = \sigma$ and $\mu = dt \mu_0$. The sign is $L_A^* \sigma + \varepsilon \Delta \sigma = \mu_0 \geq 0$. $\square$

The zero-viscosity power improvements are proved after the entropy theorem. They are unconditional consequences of the same measurable-coefficient hypotheses. The word “power” there refers to some positive exponent depending on d, λ, Λ ; no optimal exponent is asserted. More precisely, Theorem 3.1 gives a stationary weak reverse Hölder estimate on every Carnot ball, and Theorem 4.1 gives its parabolic counterpart and a uniform $L^{1+\delta}$ bound over the entire time interval. All dimensions $d \geq 1$ are covered by these positive results. The limitation on power exponents in Theorem 5.1 is proved on $\mathbb{H}^1$.

2 Proof of the entropy theorem

Throughout this section, unmarked spatial integrals are over $\mathbb{H}^d$ against dg , $|\cdot|_F$ is the Frobenius norm of a matrix, and $\nabla_W = (W_1, \dots, W_N)$ for the fields defined next.

2.1 A smooth reference family retaining the viscosity

For fixed ε , introduce the $N = 4d + 1$ divergence-free fields

$$W = (X_1, \dots, X_d, Y_1, \dots, Y_d, \sqrt{\varepsilon} \partial_{x_1}, \dots, \sqrt{\varepsilon} \partial_{y_d}, \sqrt{\varepsilon} Z).$$

The middle block denotes all $2d$ horizontal coordinate derivatives. They have constant central brackets $[W_a, W_b] = c_{ab}^\varepsilon Z$ and commute with Z . With $\Omega_0 = \begin{pmatrix} 0 & I_d \\ -I_d & 0 \end{pmatrix}$, $a = \sqrt{\varepsilon}/2$, their bracket matrix is

$$C_\varepsilon = (c_{ab}^\varepsilon) = \begin{pmatrix} \Omega_0 & a\Omega_0 & 0 \\ a\Omega_0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \|C_\varepsilon\| \leq \frac{1 + \sqrt{2}}{2}.$$

Zero fields at $\varepsilon = 0$ cause no difficulty. Set

$$\begin{aligned} L &= \sum_a W_a^2 = L_0 + \varepsilon \Delta, & L_0 &= \sum_j (X_j^2 + Y_j^2), & H_s &= e^{sL}, \\ Dv &= \sum_a W_a v_a, & \mathcal{T}R &= \sum_{a,b} W_a W_b R_{ab}. \end{aligned} \tag{2.1}$$

The stress in the original equation becomes $\text{diag}(A, I_{2d+1})\nu$, with fixed cone bounds $\min(\lambda, 1)$ and $\max(\Lambda, 1)$.

The affine diffusion for L is nonexplosive: its horizontal coordinates are Brownian linear combinations and its central coordinate is a sum of Brownian terms and linear stochastic integrals. Thus H_s is positive, conservative, and symmetric for Lebesgue measure, and commutes with central translations. At $\varepsilon = 0$, smoothing follows from the Heisenberg heat kernel displayed below; at $\varepsilon > 0$, it follows from smooth elliptic regularity. Only the following size bound is needed uniformly in ε :

$$\|H_s\|_{L^1 \rightarrow L^\infty} \leq C_d s^{-q/2}. \quad (2.2)$$

Here is a derivation of this uniformity. With Fourier convention $\hat{f}(w, \xi) = \int_{\mathbb{R}} e^{-iz\xi} f(w, z) dz$, the partial central Fourier transform of the L_0 heat kernel is

$$\hat{h}_s(w, \xi) = (4\pi s)^{-d} \left(\frac{s\xi}{\sinh(s\xi)} \right)^d \exp \left[-\frac{|w|^2}{4s} s\xi \coth(s\xi) \right], \quad w = (x, y).$$

The quotients are continued at zero. It follows by the Gaussian ansatz for $\partial_s \hat{h} = \Delta_w \hat{h} - \xi^2 |w|^2 \hat{h}/4$; the angular magnetic term annihilates radial functions of w . Fourier inversion, $u \coth u \geq 1$, and integrability of $(u/\sinh u)^d$ give $\|h_s\|_\infty \leq C_d s^{-d-1}$. Consequently

$$\|f\|_2^2 \leq C_d t^{-q/2} \|f\|_1^2 + t \sum_j (\|X_j f\|_2^2 + \|Y_j f\|_2^2).$$

Optimization is the Nash inequality for L_0 . The Dirichlet form of L dominates that of L_0 , so the same Nash inequality holds uniformly in ε . Apply it to $H_s f$ in $\frac{d}{ds} \|H_s f\|_2^2 = -2 \sum_a \|W_a H_s f\|_2^2$; the resulting differential inequality gives an $L^1 \rightarrow L^2$ bound. Symmetry and the semigroup property give (2.2). Smooth reference heat-kernel existence is used here; no measurable-coefficient heat-kernel estimate is assumed.

2.2 Exact central intertwiners

Put $J = -C_\varepsilon$ and let $\mathcal{K}B = JB - BJ$ on symmetric matrices. Both J and $\mathcal{K}$ are skew-adjoint on their respective Euclidean spaces, $\mathcal{K}I = 0$, and their norms are bounded uniformly in ε . Direct constant-bracket expansion gives

$$LD = D(L - 2ZJ), \quad L\mathcal{T} = \mathcal{T}(L - 2Z\mathcal{K}). \quad (2.3)$$

For example $[W_a, L] = 2 \sum_b c_{ab}^\varepsilon ZW_b$; applying the product rule to $[W_a W_b, L]$ and contracting with a symmetric tensor gives the second identity.

For a large fixed parameter κ , let C_s denote central convolution with

$$c_s(z) = \frac{1}{2\kappa s} \operatorname{sech} \frac{\pi z}{2\kappa s}, \quad \hat{c}_s(\xi) = \operatorname{sech}(\kappa s \xi),$$

and set

$$P_s = H_s C_s, \quad U_s = H_s C_s e^{-2sZJ}, \quad S_s = H_s C_s e^{-2sZ\mathcal{K}}. \quad (2.4)$$

The combined central factors have integrable matrix kernels. In fact, for $M = J$ or $M = \mathcal{K}$, the kernel of $C_s e^{-2sZM}$ is

$$B_s^M(z) = \frac{1}{2\kappa s} \left[\cosh \left(\frac{\pi z}{2\kappa s} I - \frac{\pi}{\kappa} M \right) \right]^{-1}.$$

To justify this formula and its bounds, diagonalize the skew-adjoint M over the complex numbers. On an eigenvalue $i\beta$ it is the analytic translate $c_s(z - 2is\beta)$, whose Fourier multiplier is $\hat{c}_s(\xi) e^{2s\beta\xi}$. This is integrable when $\kappa > 2|\beta|$. If $\kappa \geq 3\|M\|$, the elementary identity

$$\cosh(u - i\theta) = \cosh u \cos \theta - i \sinh u \sin \theta$$

gives, uniformly in u ,

$$\|B_s^M(z)\| \leq 2c_s(z), \quad \|B_s^M(z) - c_s(z)I\| \leq C\kappa^{-1}c_s(z).$$

Fourier transformation at zero shows $\int B_s^M = I$. The spectral formula also shows the kernels are real on real inputs and preserve the symmetric matrix subspace. Their masses and shrinking scales imply approximation of the identity. By positivity of H_s ,

$$|U_s F| \leq C P_s |F|, \quad |S_s R - P_s R|_F \leq C\kappa^{-1} P_s |R|_F. \quad (2.5)$$

The central difference operator, before applying H_s , commutes with every W_a . This assertion is not made for the full difference $S_s - P_s$. Exponentiating (2.3) gives the exact identities

$$DU_s = P_s D, \quad \mathcal{T}S_s = P_s \mathcal{T}. \quad (2.6)$$

One may first verify them on smooth data with compact central Fourier support and then use the integrable kernel bounds to pass to the finite measures and distributional derivatives used below.

2.3 The bracket estimate pays for central entropy

Let $|Z|$ denote the central Fourier multiplier $|\xi|$. The Heisenberg commutator gives the quadratic-form inequality

$$d\langle h, |Z|h \rangle \leq \sum_{j=1}^d (\|X_j h\|_2^2 + \|Y_j h\|_2^2) \leq \sum_a \|W_a h\|_2^2. \quad (2.7)$$

Indeed, after unitary Fourier transformation in z , put $P_j = \partial_{x_j} - i\xi y_j/2$ and $Q_j = \partial_{y_j} + i\xi x_j/2$. Then

$$\|(P_j - i \operatorname{sgn} \xi Q_j)v\|_2^2 = \|P_j v\|_2^2 + \|Q_j v\|_2^2 - |\xi| \|v\|_2^2.$$

Sum in j , integrate in ξ , and close the L^2 horizontal form. This is the use of hypoellipticity that recovers the missing central regularity in the entropy argument.

For $q \geq 0$ of mass at most one, set

$$r_s = 1 + P_s q, \quad \Phi(r) = r \log r - r + 1, \quad E(s) = \int \Phi(r_s), \quad I(s) = \int \frac{\sum_a |W_a r_s|^2}{r_s}.$$

The logarithmic derivative of $\widehat{c}_s$ is $-\kappa \xi \tanh(\kappa s \xi)$, a symmetric jump-generator symbol with nonnegative Lévy density

$$n_s(z) = \frac{\pi}{4\kappa s^2} \frac{\cosh t}{\sinh^2 t}, \quad t = \frac{\pi|z|}{2\kappa s}. \quad (2.8)$$

For instance the product expansion of cosh gives

$$\log \cosh(a\xi) = \int_{\mathbb{R}} \frac{1 - \cos(\xi z)}{2|z| \sinh(\pi|z|/(2a))} dz;$$

differentiating at $a = \kappa s$ yields (2.8). Write $J_s(r)$ for its entropy form

$$J_s(r) = \frac{1}{2} \int_{\mathbb{R}} \int_{\mathbb{R}} n_s(z) [r(g \exp(zZ)) - r(g)] [\log r(g \exp(zZ)) - \log r(g)] dg dz.$$

Scalar differentiation and integration by parts give $-E'(s) = I(s) + J_s(r_s)$. Central convolution gives $|Z \log r_s| \leq \pi/(2\kappa s)$. For positive a, b ,

$$(a - b)(\log a - \log b) \leq (4 + |\log(a/b)|)(\sqrt{a} - \sqrt{b})^2.$$

Plancherel and $\int_{\mathbb{R}} (1 - \cos(\xi v)) v^{-2} dv = \pi |\xi|$ give the central fractional-energy formula

$$\langle h, |Z|h \rangle = \frac{1}{2\pi} \int_{\mathbb{R}} \int_{\mathbb{H}^d} \frac{|h(g \exp(vZ)) - h(g)|^2}{v^2} dg dv.$$

Since $z^2 n_s(z)(4+t) \leq C\kappa$, this formula and (2.7), with $h_s = \sqrt{r_s} - 1$, imply

$$J_s(r_s) \leq C\kappa \langle h_s, |Z|h_s \rangle \leq C_d \kappa I(s).$$

There is no infinite-baseline problem: $h_s^2 \leq P_s q$, so $h_s \in L^2$. The terminal entropy is bounded by (2.2). Consequently

$$E(0) \leq C + (1 + C\kappa) \int_0^1 I(s) ds. \quad (2.9)$$

The formulas are first used for bounded integrable q , and subsequently through the uniform approximation described below.

2.4 A global estimate with a finite localization flux

First consider smooth data $q(t, g) \geq 0$, symmetric $Q(t, g)$, and a generalized source equation

$$\mathcal{T}Q = \partial_t q + \beta + DF_0 + G_0, \quad (2.10)$$

where $\beta \geq 0$, and, uniformly in t ,

$$\int q \leq 1, \quad \lambda_* q I \leq Q \leq \Lambda_* q I, \quad \int \frac{|F_0|^2}{1+q} \leq B, \quad \|G_0\|_1 \leq B.$$

Assume initially finite entropy and the regularity obtained by the presmoothing in the next subsection. Constants below may depend on B, λ_*, Λ_* but not on the entropy value or ε . Set

$$r_0 = 1 + q, \quad R_0 = I + Q, \quad r_s = P_s r_0, \quad R_s = S_s R_0, \quad a_s = R_s / r_s.$$

Choose κ so large that (2.5) gives $cr_s I \leq R_s \leq Cr_s I$, where c, C stay fixed as κ increases further. The finite coefficient perspectives for the two common scalar evolutions

$$b_s = \frac{H_s R_0}{H_s r_0}, \quad \bar{a}_s = \frac{P_s R_0}{r_s}$$

give

$$\int_0^1 \int (H_s r_0) |\nabla_W b_s|_F^2 ds \leq C, \quad \int_0^1 \int r_s |\nabla_W \bar{a}_s|_F^2 ds \leq C. \quad (2.11)$$

The perspectives used are $\int (H_s r_0) |b_s - I|^2$ and $\int r_s |\bar{a}_s - I|^2$, whose initial values are bounded by $C \int q$. For heat, differentiation gives minus twice the displayed energy; the positive central jump generator adds a nonpositive perspective term. Thus the infinite baseline never appears as an infinite variance.

Write $E_s^{\text{mat}} = R_s - P_s R_0$ and $e_s = E_s^{\text{mat}} / r_s$. For the central error operator $N_s = B_s^K * - C_s$, we have $E_s^{\text{mat}} = N_s H_s R_0$ and $W_a E_s^{\text{mat}} = N_s W_a H_s R_0$. The relative central kernel bound and weighted Cauchy–Schwarz imply

$$|e_s| \leq C/\kappa, \quad \int \frac{|W_a E_s^{\text{mat}}|^2}{r_s} \leq \frac{C}{\kappa^2} \int \frac{|W_a H_s R_0|^2}{H_s r_0}.$$

Commutation with central convolution is essential in this derivative estimate. Expand $W_a H_s R_0 = (H_s r_0) W_a b_s + b_s W_a H_s r_0$ and use the quotient rule. Since $a_s = \bar{a}_s + e_s$, (2.11) and ordinary heat entropy dissipation give

$$\int_0^1 \int r_s |\nabla_W a_s|_F^2 ds \leq C + \frac{C}{\kappa^2} E_0(t) + \frac{C}{\kappa^2} \int_0^1 I(s, t) ds, \quad E_0(t) = \int \Phi(1 + q(t)). \quad (2.12)$$

Here the ordinary heat Fisher integral is at most $E_0(t)$. The constants in (2.12) do not grow with κ .

By (2.6), equation (2.10) becomes

$$\mathcal{T}R_s = \partial_t r_s + P_s \beta + DU_s F_0 + P_s G_0.$$

Weighted kernel Jensen gives $\int |U_s F_0|^2 / r_s \leq CB$. Test with the nonnegative $\log r_s$ and integrate once in the horizontal fields. The principal term is $-\int r_s a_{s,ab} (W_a \log r_s) (W_b \log r_s)$; the coefficient cross term costs a small fraction of I and $C \int r_s |\nabla_W a_s|^2$. The positive source has the favorable sign. The signed scalar source costs at most $CB(1 + |\log s|)$, since (2.2) gives

$$\|\log r_s(t)\|_\infty \leq C(1 + |\log s|), \quad E(s, t) \leq C(1 + |\log s|). \quad (2.13)$$

Thus, for $0 \leq \theta \leq 1$, $\theta \in C_c^\infty(0, T)$, temporal integration by parts yields

$$\int \theta(t) I(s, t) dt \leq C \int \theta(t) \int r_s |\nabla_W a_s|^2 dt + C(T + \|\theta'\|_1)(1 + |\log s|). \quad (2.14)$$

The temporal contribution is $\int \theta' E(s, t)$, controlled by (2.13); no value of the entropy at $t = 0$ or T is used.

All global tests can be made first with Carnot cutoffs tending to one. For radius $R \geq 1$ their first and second W derivatives are bounded by C/R and C/R^2 uniformly in ε . For almost every positive regularization parameter s , the weighted coefficient, Fisher, and flux quantities are finite by their integrated bounds; perform the test at these s and then integrate in s . Moreover, $(\log r_s)^2 r_s \leq C(s)(r_s - 1)$, and the scalar source is integrable. Subtract the constant tensor before integrating by parts; it has zero double divergence. Cauchy–Schwarz removes the first-order cutoff terms, and the finite perturbation removes the second-order terms. The symmetric jump identities follow by truncating their kernels, using convexity, and passing by monotone convergence or finite perturbation domination. These arguments need no uniform bounds on heat-kernel derivatives.

Put $\mathbf{E} = \int \theta E_0$ and $\mathbf{I} = \int \theta(t) \int_0^1 I(s, t) ds dt$. Before absorption these are finite for the presmoothed data; in particular the positive scalar entropy identity gives $\mathbf{I} \leq \mathbf{E}$. Integrating (2.12) and (2.14), then absorbing $C\kappa^{-2}\mathbf{I}$, gives

$$\mathbf{I} \leq C(T + \|\theta'\|_1) + C\kappa^{-2}\mathbf{E}.$$

On the other hand (2.9) gives $\mathbf{E} \leq CT + (1 + C\kappa)\mathbf{I}$. A second choice of sufficiently large fixed κ absorbs the coefficient $C(1 + C\kappa)/\kappa^2$ of $\mathbf{E}$. Therefore

$$\int \theta(t) \int \Phi(1 + q(t, g)) dg dt \leq C(T + \|\theta'\|_1). \quad (2.15)$$

This is the complete entropy closure. Quadratic smallness of the tensor error and linear growth of the central entropy cost are both used.

2.5 Localization, rough measures and endpoints

Return to the measures in Theorem 1.1. Fix $\eta \in C_c^\infty(\Omega)$, $0 \leq \eta \leq 1$, equal to one near K , and put $\chi = \eta^2$. On all space define

$$q = \chi \nu, \quad Q = \chi \operatorname{diag}(A, I_{2d+1}) \nu,$$

extending the finite measures by zero. The product rule, with $\widehat{Q} = \operatorname{diag}(A, I_{2d+1}) \nu$, gives exactly

$$\mathcal{T}Q = \partial_t q + \beta + DF_0 + G_0, \quad \beta = \chi \mu \geq 0,$$

$$(F_0)_a = 2 \sum_b (W_b \chi) \hat{Q}_{ab}, \quad G_0 = - \sum_{a,b} (W_a W_b \chi) \hat{Q}_{ab}.$$

Symmetry justifies the derivative ordering. On each time slice let $r_{0,t} = dg + q_t$. These flux measures satisfy

$$\int \left| \frac{dF_{0,t}}{dr_{0,t}} \right|^2 dr_{0,t} \leq C, \quad |G_{0,t}|(\mathbb{H}^d) \leq C. \quad (2.16)$$

Indeed $|\nabla_W \chi|^2 / \chi \leq 4|\nabla_W \eta|^2$ on $\chi > 0$, while $W\chi = 0$ where $\chi = 0$. All cutoff constants are uniform in ε . No absolute continuity of ν is needed for this Radon–Nikodym calculation.

An ordinary group mollifier need not commute with the coordinate viscosity. Instead choose a fixed sufficiently large presmoothing parameter κ_{pre} and apply the exact regularizers to these already localized measures:

$$q_\delta = P_\delta q, \quad Q_\delta = S_\delta Q, \quad \beta_\delta = P_\delta \beta, \quad F_{0,\delta} = U_\delta F_0, \quad G_{0,\delta} = P_\delta G_0.$$

Their equation is unchanged in form. The scalar mass is at most one, the matrix cone is enlarged by fixed factors only, $\beta_\delta \geq 0$, and (2.16) survives by positive-kernel weighted Jensen and (2.5). The scalar source keeps its L^1 bound. The tensor cone comparison is with q_δ itself; no artificial finite density is inserted into the target measure. The baseline is introduced only in the entropy proof.

Convolve also in time with a positive compact smooth kernel. Extend the localized slice measures by zero outside $(0, T)$ for this convolution, but use the equation only on interior time slabs whose distance from the endpoints exceeds the convolution radius. There are consequently no endpoint source terms. For each fixed compactly supported θ , the resulting triples are smooth, have finite entropy by (2.2), retain all the preceding uniform bounds, and satisfy the generalized equation on its support. Apply (2.15), with a second large parameter chosen for the already enlarged cone. The constants are independent of both smoothing radii and of ε .

The bound is superlinear in the scalar density. On compact space-time sets, uniform integrability and weak L^1 compactness identify the measure limit $\chi\nu$ with a density. Both scalar regularizers are approximate identities, so the weak measure limit is the original $\chi\nu$. Convex lower semicontinuity gives the same entropy bound. Exhausting the spatial domain and interior time interval identifies a single density p of ν . Taking cutoffs $\theta_j \rightarrow 1$ on $(0, T)$ with $0 \leq \theta_j \leq 1$ and $\|\theta'_j\|_1 \leq 2$, and using Fatou, yields the estimate over the entire time interval. Finally,

$$p \log(1 + p) \leq \Phi(1 + p) + p,$$

and the total mass equals $\nu((0, T) \times \Omega) \leq T$. This proves Theorem 1.1. $\square$

3 A stationary power gain at zero viscosity

Let d_{CC} be the left-invariant Carnot–Carathéodory distance for the displayed Heisenberg fields, write $B(g, r)$ for its open balls, and set $Q = 2d + 2$. Haar measure satisfies $|B(g, r)| = v_d r^Q$, where $v_d = |B(e, 1)|$. For $c > 0$, the notation cB means the concentric ball with radius multiplied by c .

Theorem 3.1 (Stationary weak reverse Hölder estimate). *For every $d \geq 1$ and $0 < \lambda \leq \Lambda < \infty$ there are $\delta > 0$ and $C < \infty$, depending only on d, λ, Λ , with the following property. Let $\Omega \subset \mathbb{H}^d$ be open, let σ, μ be nonnegative locally finite Borel measures on Ω , and let A be Borel and symmetric with $\lambda I \leq A \leq \Lambda I$ σ -almost everywhere. Suppose*

$$\int L_A \varphi d\sigma = \int \varphi d\mu \quad (\varphi \in C_c^\infty(\Omega)).$$

Then $\sigma = p dx$, $p \in L_{\text{loc}}^{1+\delta}(\Omega)$, and, for every $B = B(g, r)$ with $\overline{2B} \subset \Omega$,

$$\left(\int_B p^{1+\delta} \right)^{1/(1+\delta)} \leq C \int_{2B} p. \quad (3.1)$$

In particular, for every Borel $E \subset B$,

$$\sigma(E) \leq C \left(\frac{|E|}{|B|} \right)^{\delta/(1+\delta)} \sigma(2B). \quad (3.2)$$

No doubling assumption on σ and no bound on $\mu(2B)$ is imposed. The equation in this theorem has $\varepsilon = 0$.

We first extract the scale-invariant quantitative input from Corollary 1.2, and then prove its self-improvement.

Lemma 3.2 (Relative entropy on every ball). *Under the assumptions of Theorem 3.1, $\sigma = p dx$ and there is $H = H(d, \lambda, \Lambda) < \infty$ such that, whenever $\overline{2B} \subset \Omega$ and $m = \sigma(2B) > 0$,*

$$\int_B p \log \left(1 + \frac{|B|p}{m} \right) \leq Hm. \quad (3.3)$$

Consequently, for $E \subset B$ Borel,

$$\sigma(E) \leq \omega \left(\frac{|E|}{|B|} \right) \sigma(2B), \quad \omega(s) = \min \left\{ 1, \sqrt{s} + \frac{H}{\log(1 + s^{-1/2})} \right\} \quad (0 < s \leq 1), \quad (3.4)$$

with $\omega(0) = 0$. In particular ω is nondecreasing and tends to zero at zero.

Proof. Local absolute continuity follows from Corollary 1.2: restrict to any relatively compact ball and divide the finite mass by a positive constant to make it at most one. Such balls cover Ω . If $m = 0$, all assertions on B are immediate. Otherwise put $F(y) = g\delta_r y$, where $B = B(g, r)$, and define on $B(e, 2)$

$$\tilde{\sigma} = m^{-1}(F^{-1})_{\#}(\sigma|_{2B}), \quad \tilde{A} = A \circ F, \quad \tilde{\mu} = \frac{r^2}{m}(F^{-1})_{\#}(\mu|_{2B}).$$

Left invariance and degree-one homogeneity of every horizontal field give

$$L_A(\varphi \circ F^{-1})(F(y)) = r^{-2} L_{\tilde{A}} \varphi(y).$$

Thus $L_{\tilde{A}}^* \tilde{\sigma} = \tilde{\mu} \geq 0$, $\tilde{\sigma}(B(e, 2)) = 1$, and the ellipticity bounds are unchanged. The source is finite because $\overline{2B} \subset \Omega$; its total mass is not bounded or used. The density is $\tilde{p}(y) = r^Q p(F(y))/m$. Apply Corollary 1.2 with the fixed domain $B(e, 2)$ and compact set $\overline{B(e, 1)}$. Its constant is independent of g, r, m, A and μ . Since

$$\log(1 + v_d t) \leq \log(1 + t) + \log \max\{1, v_d\},$$

the resulting entropy bound is exactly (3.3), with a possibly enlarged constant H .

Put $u = |B|p/m$. Then $\int_B u \log(1 + u) \leq H$. For $s = |E|/|B| > 0$, split E according to $u \leq s^{-1/2}$ or $u > s^{-1/2}$. These two parts contribute at most $\sqrt{s}$ and $H/\log(1 + s^{-1/2})$, respectively, to $\sigma(E)/m$. The elementary bound $\sigma(E) \leq m$ gives the minimum in (3.4). Null sets have zero σ -mass by absolute continuity. $\square$

The following lemma includes both the Heisenberg metric and the product metric $\max\{d_{\text{CC}}(x, y), |t - s|^{1/2}\}$ on $\mathbb{H}^d \times \mathbb{R}$. Their volume exponents are Q and $Q + 2$, respectively. Stating it in this form separates the geometric argument from the differential equation.

Lemma 3.3 (A small-set modulus implies a weak reverse Hölder bound). *Let $(\mathcal{X}, d, \mathbf{m})$ be a proper metric space with a Radon measure of full support, and suppose every open ball satisfies $\mathbf{m}(B(x, r)) = vr^q$ for fixed $v > 0$ and $q > 0$. Fix $\tau > 1$, $\kappa > 1$, a ball B_0 , and a nonnegative $w \in L^1(\kappa B_0, \mathbf{m})$. Suppose a nondecreasing function $\omega : [0, 1] \rightarrow [0, \infty)$ with $\omega(s) \rightarrow 0$ as $s \downarrow 0$ satisfies*

$$\int_E w \, d\mathbf{m} \leq \omega\left(\frac{\mathbf{m}(E)}{\mathbf{m}(D)}\right) \int_{\tau D} w \, d\mathbf{m} \quad (3.5)$$

for every ball D with $\overline{\tau D} \subset \kappa B_0$ and every Borel $E \subset D$. There are $\theta > 0$ and, for every $0 < s < \theta$, a constant C_s , depending only on $q, \tau, \kappa, \omega, s$, such that

$$\left(\int_{B_0} w^{1+s} \, d\mathbf{m} \right)^{1/(1+s)} \leq C_s \int_{\kappa B_0} w \, d\mathbf{m}. \quad (3.6)$$

Only the volume measure $\mathbf{m}$ is doubling; the measure $w \, d\mathbf{m}$ need not be doubling.

Proof. Write $B_0 = B(x_0, r_0)$ and $M = \int_{\kappa B_0} w \, d\mathbf{m}$. The case $M = 0$ is immediate. Extend $f = w \mathbf{1}_{B_0}$ by zero to $\mathcal{X}$, and let $\mathcal{M}f$ be the uncentered maximal function over open metric balls. The usual disjoint-ball selection gives

$$\mathbf{m}\{\mathcal{M}f > a\} \leq \frac{5^q}{a} \|f\|_1. \quad (3.7)$$

Here is the relevant geometric content. From a family of balls of bounded radii one selects disjoint balls D_j whose fivefold dilates cover its union, by selecting maximal disjoint collections in decreasing dyadic radius ranges. Apply this to balls with f -average greater than a ; their radii are bounded by $(\|f\|_1/(va))^{1/q}$, and $\sum_j \mathbf{m}(D_j) \leq \|f\|_1/a$. The exact volume law gives (3.7). The same bound, approximation of L^1 functions by continuous functions, and continuity on shrinking balls prove the Lebesgue differentiation theorem for $\mathbf{m}$. In particular $w \leq \mathcal{M}f$ almost everywhere on B_0 .

Put $O_a = \{\mathcal{M}f > a\}$, which is open because the maximal function is uncentered. If $x \notin \kappa' B_0$, where $\kappa' = (1 + \kappa)/2$, a ball containing x and meeting B_0 has radius at least $(\kappa - 1)r_0/4$. Its f -average is consequently at most $(4/(\kappa - 1))^q M/\mathbf{m}(B_0)$. Choose a fixed $c_0 > (4/(\kappa - 1))^q$ and set

$$a_0 = c_0 M/\mathbf{m}(B_0).$$

For $a \geq a_0$ it follows that $\overline{O_a} \subset \overline{\kappa' B_0} \subset \kappa B_0$; the first closed set is compact by properness.

We now describe a Whitney covering of each nonempty O_a . Let $\rho(x) = d(x, O_a^c)$ and choose a fixed $L > 100(1 + \tau)$. Apply the disjoint-ball selection to $b_x = B(x, \rho(x)/L)$. We obtain pairwise disjoint balls b_j with $D_j = 5b_j$ covering O_a . If $D_j = B(x_j, r_j)$, then $r_j = 5\rho(x_j)/L$ and

$$\overline{\tau D_j} \subset O_a.$$

Moreover these dilates have bounded overlap, say at most $N = N(q, \tau)$. To verify this, if $y \in \tau D_i \cap \tau D_j$, the one-Lipschitz property of ρ gives

$$|\rho(x_i) - \rho(x_j)| \leq \frac{5\tau}{L} (\rho(x_i) + \rho(x_j)).$$

Thus all radii of dilates through one point are comparable by a fixed factor. Their disjoint inner balls b_j lie in one ball with radius a fixed multiple of any of these radii. Comparing their volumes vr^q bounds their number by N . This is a statement about volume alone.

Properness supplies a point $z_j \in O_a^c$ with $d(x_j, z_j) = \rho(x_j) = (L/5)r_j$. Since $\mathcal{M}f(z_j) \leq a$, a ball centered at x_j of radius $(L/5 + 4)r_j$ gives

$$\int_{3D_j} f \, d\mathbf{m} \leq C_1 a \mathbf{m}(D_j), \quad C_1 = C_1(q, \tau). \quad (3.8)$$

If an averaging ball of radius at least r_j contains a point of D_j , enlarging it by the fixed factor $L/5+3$ makes it contain z_j . Its average is therefore at most $A_0 a$, for a fixed $A_0 = A_0(q, \tau) > 1$. If $A > A_0$, a ball witnessing $\mathcal{M}f > Aa$ at a point of D_j must have radius less than r_j ; it is then contained in $3D_j$. By (3.7) applied to $f\mathbf{1}_{3D_j}$ and (3.8),

$$\frac{\mathbf{m}(O_{Aa} \cap D_j)}{\mathbf{m}(D_j)} \leq \frac{C_2}{A}, \quad C_2 = C_2(q, \tau).$$

Choose $A > \max\{A_0, C_2, 1\}$ so large that $N\omega(C_2/A) \leq 1/2$. Apply (3.5) to $O_{Aa} \cap D_j$, sum over the covering, and use the bounded overlap of $\tau D_j \subset O_a$. This gives the weighted level-set decay

$$\int_{O_{Aa}} w \, d\mathbf{m} \leq \frac{1}{2} \int_{O_a} w \, d\mathbf{m} \quad (a \geq a_0). \quad (3.9)$$

The integrals here use the original w throughout O_a , including any part outside B_0 . They are bounded by the same outer mass M .

Set $\theta = \log 2 / \log A$. Iteration of (3.9) and differentiation on B_0 yield, for $R \geq a_0$,

$$\int_{\{x \in B_0 : w(x) > R\}} w \, d\mathbf{m} \leq 2M(R/a_0)^{-\theta}.$$

Tonelli's theorem applied to the weighted layer-cake formula now gives, for $0 < s < \theta$,

$$\int_{B_0} w^{1+s} \, d\mathbf{m} \leq \left(1 + \frac{2s}{\theta - s}\right) M a_0^s. \quad (3.10)$$

Since $a_0 = c_0 M / \mathbf{m}(B_0)$ and $\mathbf{m}(\kappa B_0) = \kappa^q \mathbf{m}(B_0)$, this is (3.6). At no step was a weight-doubling inequality used. $\square$

Proof of Theorem 3.1. Apply Lemma 3.3 to $w = p$ on $2B$, with $q = Q$, $\tau = \kappa = 2$, and the modulus in Lemma 3.2. Every Whitney testing double is compactly contained in $2B$, so its relative entropy estimate follows by restricting the same equation. Taking, for example, $\delta = \theta/2$ proves (3.1). Hölder's inequality on $E \subset B$ gives (3.2), after changing the constant. $\square$

Proposition 3.4 (Coordinate viscosity prevents a uniform Carnot-ball estimate). *For every $d \geq 1$ and every $s > 0$, estimate (3.1) with exponent $1 + s$ cannot hold with one constant on all Carnot balls for the class $L_A^* \sigma + \Delta_{\text{coord}} \sigma = \mu \geq 0$, even when $A = I$, $\Omega = B(e, 1)$, and both measures have smooth densities with $\sigma(\Omega) \leq 1$ and $\mu(\Omega) < \infty$.*

Proof. Write $h = (x, y) \in \mathbb{R}^n$, $n = 2d$, and for $a > 0$ set

$$p_a(h, z) = \exp\left(-\frac{|h|^2}{2a^2}\right) \cosh\left(\frac{\sqrt{2n}z}{a}\right).$$

For functions radial in h , the rotational cross term in the Heisenberg sublaplacian vanishes. Direct differentiation therefore gives

$$\frac{(L_I + \Delta_{\text{coord}})p_a}{p_a} = 2\left(\frac{|h|^2}{a^4} - \frac{n}{a^2}\right) + \left(1 + \frac{|h|^2}{4}\right) \frac{2n}{a^2} = \frac{2|h|^2}{a^4} + \frac{n|h|^2}{2a^2} \geq 0.$$

The operators here are self-adjoint, so these are admissible positive smooth densities with a nonnegative smooth source on Ω .

Take $a = r^2$ and let $r \downarrow 0$. Elementary horizontal paths give

$$\{|h| < r/4, |z| < r^2/64\} \subset B(e, r) \subset \{|h| < r, |z| < r^2/2\}.$$

For the first inclusion, make a horizontal square loop of length $4\sqrt{|z|}$ producing central displacement z , and then a straight horizontal segment of length $|h|$. For the second, along a horizontal path of length less than r one has $|h| < r$ and $|z| \leq \frac{1}{2} \int |h| |\dot{h}| < r^2/2$. On $B(e, 2r)$ the hyperbolic cosine in $p_{r,2}$ is bounded by a constant depending only on d . Gaussian integration and the inner box thus give

$$\int_{B(e, 2r)} p_{r,2} \leq C_d r^{2n+2}, \quad \int_{B(e, r)} p_{r,2}^{1+s} \geq c_{d,s} r^{2n+2}.$$

Because $|B(e, r)| = v_d r^{n+2}$, their reverse Hölder ratio is at least $c_{d,s} r^{-ns/(1+s)}$, which tends to infinity. Divide each density by its finite integral on Ω to give mass one; the ratio is unchanged, and its smooth nonnegative source remains finite. This proves the claim. $\square$

Remark 3.5. The obstruction concerns the scale-invariant estimate on fixed Carnot balls. It does not contradict the local entropy estimate, uniform in $0 \leq \varepsilon \leq 1$, in Corollary 1.2. Under a Carnot dilation the central coordinate viscosity has coefficient ε/r^2 after normalizing the horizontal operator; left translation also shears the coordinate directions. Thus that viscous class does not have the exact covariance used in Lemma 3.2.

4 A parabolic power gain at zero viscosity

The stationary improvement has a space-time counterpart. The slice mass bound in Theorem 1.1 also allows the gain to be integrated all the way to both time endpoints.

For $a = (t_0, g_0)$ define the parabolic balls

$$\mathcal{Q}_r(a) = (t_0 - r^2, t_0 + r^2) \times B(g_0, r).$$

They are the balls of the proper metric

$$d_P((t, g), (s, h)) = \max\{|t - s|^{1/2}, d_{CC}(g, h)\},$$

and $|\mathcal{Q}_r| = 2v_d r^{Q+2}$, where $Q = 2d + 2$ as above.

Theorem 4.1 (Parabolic higher integrability). *Under exactly the hypotheses of Theorem 1.1, assume in addition that $\varepsilon = 0$. There are $\delta_0 > 0$ and $C_0 < \infty$, depending only on d, λ, Λ , such that, for every parabolic ball with $\overline{\mathcal{Q}_{2r}(a)} \subset (0, T) \times \Omega$,*

$$\left(\int_{\mathcal{Q}_r(a)} p^{1+\delta_0} \right)^{1/(1+\delta_0)} \leq C_0 \int_{\mathcal{Q}_{2r}(a)} p. \quad (4.1)$$

Moreover, for every

$$0 < \delta < \min\{\delta_0, 2/Q\}$$

and every $K \Subset \Omega$,

$$\int_0^T \int_K p^{1+\delta} dg dt \leq C(T, K, \Omega, d, \lambda, \Lambda, \delta). \quad (4.2)$$

In particular, one may choose a positive δ depending only on d, λ, Λ . Neither estimate requires an initial density or initial entropy bound, and both constants are independent of A, ν, μ .

Lemma 4.2 (Normalization on a parabolic ball). *Under these assumptions there is a nondecreasing modulus $\omega_P(s) \rightarrow 0$ as $s \downarrow 0$, depending only on d, λ, Λ , such that*

$$\nu(E) \leq \omega_P \left(\frac{|E|}{|\mathcal{Q}_r(a)|} \right) \nu(\mathcal{Q}_{4r}(a)) \quad (4.3)$$

whenever $\overline{\mathcal{Q}_{4r}(a)} \subset (0, T) \times \Omega$ and $E \subset \mathcal{Q}_r(a)$ is Borel.

Proof. By Theorem 1.1 we already have a density. Write $M = \nu(\mathcal{Q}_{4r}(a))$; the case $M = 0$ is immediate. Choose a fixed smooth cutoff on the group, equal to one on $B(e, 2)$, supported in $B(e, 3)$, and taking values in $[0, 1]$. Translation and dilation give a cutoff χ equal to one on $B(g_0, 2r)$, supported in $B(g_0, 3r)$, and satisfying

$$|L_A \chi| \leq Cr^{-2}$$

on this support for ν -almost every point. The constant uses no derivatives of A . Put $F(t) = \int \chi d\nu_t$. Testing the equation with products of a time test and χ shows, on $(t_0 - 16r^2, t_0 + 16r^2)$, that

$$F' = G(t) dt - \mu_\chi, \quad G(t) = \int L_A \chi d\nu_t, \quad \mu_\chi \geq 0.$$

Here μ_χ is the time projection of $\chi\mu$. Thus F has a locally bounded-variation representative, and for almost every pair of Lebesgue times $s < t$,

$$F(t) \leq F(s) + Cr^{-2} \int_s^t \nu_\tau(B(g_0, 3r)) d\tau. \quad (4.4)$$

Averaging on the past interval $(t_0 - 15r^2, t_0 - 9r^2)$ supplies a Lebesgue time s with $F(s) \leq M/(6r^2)$ for which (4.4) holds for almost every later time. Consequently a fixed $c_* > 0$ gives

$$\nu_t(B(g_0, 2r)) \leq c_* M/r^2 \quad \text{for almost every } t \in (t_0 - 4r^2, t_0 + 4r^2). \quad (4.5)$$

The sink sign is exactly what permits this estimate without a source-mass term.

Set $F_r(h) = g_0 \delta_r h$ and introduce, for $-4 < s < 4$,

$$\tilde{\nu}_s = \frac{r^2}{c_* M} (F_r^{-1})_\# (\nu_{t_0 + r^2 s} |_{B(g_0, 2r)}).$$

These measures have mass at most one on $B(e, 2)$ by (4.5). Their space-time density is

$$\tilde{p}(s, h) = \frac{r^{Q+2}}{c_* M} p(t_0 + r^2 s, F_r(h)).$$

The transformed matrix is $\tilde{A}(s, h) = A(t_0 + r^2 s, F_r(h))$. Writing $\mathcal{F}(s, h) = (t_0 + r^2 s, F_r(h))$, the transformed source is

$$\tilde{\mu} = \frac{r^2}{c_* M} (\mathcal{F}^{-1})_\# (\mu|_{(t_0 - 4r^2, t_0 + 4r^2) \times B(g_0, 2r)}).$$

It is finite and nonnegative. The horizontal homogeneity, together with the time scaling, verifies the same weak parabolic equation with $\varepsilon = 0$ on $(-4, 4) \times B(e, 2)$, and the same ellipticity constants. Applying Theorem 1.1 there (after a time translation) gives

$$\int_{\mathcal{Q}_1(0, e)} \tilde{p} \log(1 + \tilde{p}) \leq C.$$

Changing variables and adjusting the fixed factor $2v_d c_*$ yields

$$\int_{\mathcal{Q}_r(a)} p \log \left(1 + \frac{|\mathcal{Q}_r(a)|p}{M} \right) \leq HM, \quad (4.6)$$

where H is universal for the stated parameters. In this adjustment we use $\nu(\mathcal{Q}_r(a)) \leq M$ and $\log(1 + bu) \leq \log(1 + u) + \log \max\{1, b\}$. Splitting at the level $u = s^{-1/2}$ with $u = |\mathcal{Q}_r(a)|p/M$ proves (4.3), with

$$\omega_P(s) = \min \left\{ 1, \sqrt{s} + \frac{H}{\log(1 + s^{-1/2})} \right\} \quad (0 < s \leq 1), \quad \omega_P(0) = 0.$$

□

Proof of Theorem 4.1. Apply Lemma 3.3 in the parabolic metric with volume exponent $Q+2$, input dilation $\tau = 4$, and outer dilation $\kappa = 2$. Every testing ball whose fourfold dilation is compactly contained in $\mathcal{Q}_{2r}(a)$ satisfies Lemma 4.2. Choosing a fixed exponent below the lemma's threshold proves (4.1), with constants depending only on d, λ, Λ .

We give the endpoint summation explicitly. Fix $K \Subset \Omega$ and choose $\rho > 0$ such that $B(g, 4\rho) \Subset \Omega$ for every $g \in K$. For $j \geq 1$ put

$$a_j = T2^{-j-1}, \quad J_j = (a_j, 2a_j), \quad r_j = \min\{\rho, \sqrt{a_j}/8\}.$$

The intervals J_j cover $(0, T/2)$ up to endpoints; their reflections $T - J_j$ cover $(T/2, T)$. Cover K by balls of radius r_j with centers in K whose doubled balls have bounded overlap, uniformly in j . Such a cover follows by taking a maximal r_j -separated set and using the exact ball volume law. Cover J_j by time intervals of radius r_j^2 , centered in its closure, whose intervals of radius $4r_j^2$ have bounded overlap. The product cover consists of balls $\mathcal{Q}_{r_j}(a_{j\ell})$, and the doubled parabolic balls have bounded overlap. They are compactly contained in $(0, T) \times \Omega$. Their times lie in an interval of length at most Ca_j inside $(0, T)$. The same assertions hold for the reflected strip.

For $0 < \delta < \delta_0$, Hölder's inequality first lowers the exponent in (4.1) to $1 + \delta$. If $M_{j\ell} = \nu(\mathcal{Q}_{2r_j}(a_{j\ell}))$, this gives

$$\int_{\mathcal{Q}_{r_j}(a_{j\ell})} p^{1+\delta} \leq Cr_j^{-(Q+2)\delta} M_{j\ell}^{1+\delta}.$$

The original slice mass bound implies

$$M_{j\ell} \leq 8r_j^2, \quad \sum_{\ell} M_{j\ell} \leq Ca_j.$$

Therefore, for either endpoint strip,

$$\int_{J_j} \int_K p^{1+\delta} \leq Ca_j r_j^{-Q\delta}, \tag{4.7}$$

with J_j replaced by $T - J_j$ when appropriate. Only finitely many j have $r_j = \rho$ instead of $r_j = \sqrt{a_j}/8$. The tail of the series in (4.7) is bounded by a constant times

$$\sum_j a_j^{1-Q\delta/2},$$

which converges for $\delta < 2/Q$. This proves (4.2), with the asserted dependencies. For example $\delta = \min\{\delta_0/2, 1/Q\}$ is an admissible choice depending only on d, λ, Λ . $\square$

Remark 4.3. The restriction $\delta < 2/Q$ in the full-time conclusion arises from the summation near a possible singular initial measure. The interior estimate (4.1) is stated with its independent exponent δ_0 . No claim of optimality of either exponent is made. The power theorems in this note concern $\varepsilon = 0$; the uniform conclusion for the entire interval $0 \leq \varepsilon \leq 1$ is the entropy theorem.

5 A limit on possible higher-integrability exponents

An entropy or small-power estimate does not extend to all finite powers, even on the first Heisenberg group. We prove this with one exact stationary family, allowing every prescribed ellipticity ratio greater than one. Its integrability threshold is explicit; no assertion that this threshold is optimal for the entire class is needed or made.

In this section

$$X = \partial_x - \frac{y}{2}\partial_z, \quad Y = \partial_y + \frac{x}{2}\partial_z, \quad dg = dx dy dz,$$

and

$$\mathcal{L}_A \varphi = a_{11}X^2\varphi + a_{12}(XY + YX)\varphi + a_{22}Y^2\varphi.$$

Stationarity means $\int p \mathcal{L}_A \varphi dg = 0$ for every compactly supported smooth test. Thus both the sink and the additional Euclidean viscosity are zero throughout this construction.

Theorem 5.1 (Exact obstruction at every contrast). *Fix $0 < \lambda < \Lambda < \infty$, write $L = \Lambda/\lambda$, and set*

$$\Omega_0 = \{x^2 + y^2 < 1, |z| < 1\}, \quad K_0 = \{x^2 + y^2 \leq 1/4, |z| \leq 1/2\}, \quad s_* = \frac{4L}{3(L-1)}.$$

There exist a nonnegative probability density $p \in L^1(\Omega_0)$ and a Borel symmetric matrix A , with $\lambda I \leq A \leq \Lambda I$, such that

$$\mathcal{L}_A^* p = 0 \quad \text{in } \Omega_0, \quad p \notin L^{s_*}(K_0).$$

There also exist smooth strictly positive probability densities p_j and smooth symmetric matrices A_j , with the same bounds, such that

$$\mathcal{L}_{A_j}^* p_j = 0, \quad p_j \longrightarrow p \text{ in } L^1(\Omega_0), \quad \int_{K_0} p_j^{s_*} dg \longrightarrow \infty. \quad (5.1)$$

For every fixed $1 < s < s_$,*

$$p \in L^s(\Omega_0), \quad \sup_j \int_{\Omega_0} p_j^s dg < \infty. \quad (5.2)$$

Consequently this family has uniformly bounded entropy. At $L = 3$, it excludes even qualitative local L^2 membership, as well as uniform local L^2 bounds for smooth stationary densities.

Proof. *The density and stress.* Put

$$\zeta = x + iy = re^{i\theta}, \quad u = r^2/4, \quad D = u - iz,$$

and choose

$$\eta = \frac{L-1}{2}, \quad \kappa = \frac{\eta}{\eta+1}, \quad \beta = \frac{\eta(\eta+2)}{2\eta+1}, \quad \alpha = 4\beta - 2\eta = \frac{6\eta}{2\eta+1} < 3. \quad (5.3)$$

For $r > 0$ define

$$q = u^\eta(u^2 + z^2)^{-\beta}, \quad w = \kappa q e^{-2i\theta} \frac{D}{\overline{D}}, \quad Q = \begin{pmatrix} q + \operatorname{Re} w & -\operatorname{Im} w \\ -\operatorname{Im} w & q - \operatorname{Re} w \end{pmatrix}. \quad (5.4)$$

Give q, Q the value zero on the vertical axis, a null set. Since $|w| = \kappa q$, the eigenvalues of Q are $(1 - \kappa)q$ and $(1 + \kappa)q$. With $c = (\lambda + \Lambda)/2$, the matrix $A = cQ/q$ therefore has eigenvalues exactly λ, Λ off the axis; set $A = cI$ on the axis.

Let $Z = X + iY$, $\Delta_H = X^2 + Y^2$, and

$$\mathcal{T}Q = X^2Q_{11} + (XY + YX)Q_{12} + Y^2Q_{22}.$$

The fields are divergence free and

$$\mathcal{T}Q = \Delta_H q + \operatorname{Re}(Z^2 w). \quad (5.5)$$

Here the minus sign in the off-diagonal entry of Q is essential. We verify cancellation explicitly. Write $v = u^2 + z^2$. On radial functions, $\Delta_H = \partial_u + u(\partial_{uu} + \partial_{zz})$, so

$$\frac{\Delta_H q}{q} = \frac{\eta^2}{u} + 2\beta(2\beta - 2\eta - 1)\frac{u}{v}.$$

For $P = \partial_u + i\partial_z$ and any radial f ,

$$Z^2(\bar{\zeta}^2 f) = 8f + 16uPf + 4u^2P^2f.$$

Substitute $f = (\kappa/4)u^{\eta-1}D^{1-\beta}\bar{D}^{-1-\beta}$, so that $w = \bar{\zeta}^2 f$. Differentiation gives

$$\frac{\text{Re}(Z^2 w)}{q} = \kappa \left\{ \frac{\eta(\eta+1)(u^2 - z^2)}{uv} + 4(\beta-1)(\beta-\eta-1)\frac{u}{v} \right\}.$$

Using $\kappa = \eta/(\eta+1)$, their sum factors as

$$\frac{\mathcal{T}Q}{q} = \frac{4(2\eta+1)}{\eta+1} \left(\beta - \frac{\eta(\eta+2)}{2\eta+1} \right) \left(\beta - \frac{\eta+1}{2} \right) \frac{u}{v} = 0. \quad (5.6)$$

Thus the equation holds pointwise off the axis.

The equation across the singular set. Both q and Q are homogeneous of degree $-\alpha$ under $\delta_a(x, y, z) = (ax, ay, a^2z)$, whose volume factor is a^4 . They are locally integrable: on a compact intrinsic annulus the functions are bounded, including near the axis, and $\alpha < 4$. Let $\mathfrak{D} = \mathcal{T}Q$ as a distribution. For a test φ supported away from the origin, multiply by a smooth horizontal cutoff equal to one on $r < a$ and zero on $r > 2a$. Since $\mathfrak{D}$ vanishes off the axis and $|Q| \leq Cr^{2\eta}$ near the axis on this support,

$$|\langle \mathfrak{D}, \varphi \rangle| \leq Ca^{-2} \int_{r < 2a} r^{2\eta} dx dy dz \leq Ca^{2\eta} \rightarrow 0.$$

Hence $\mathfrak{D}$ is supported at the origin.

If $\varphi(0) = 0$, use a smooth cutoff equal to one near the origin and supported at intrinsic radius $2a$, obtained from a fixed cutoff by $\delta_{1/a}$. On its support $|\varphi| \leq Ca$; the horizontal second derivatives of the product are bounded by C/a . Homogeneity gives

$$|\langle \mathfrak{D}, \varphi \rangle| \leq Ca^{-1}a^{4-\alpha} = Ca^{3-\alpha} \rightarrow 0.$$

It follows directly that $\mathfrak{D} = b\delta_0$ for a constant b : its value on a test depends only on the test's value at zero.

It remains to compute b ; pointwise cancellation alone would not justify discarding it. Take $\varphi(x, y, z) = \psi(v)$, with $\psi \in C_c^\infty(\mathbb{R})$ equal to one near zero. The symmetrized horizontal Hessian of v is $3uI$, and $|\nabla_H v|^2 = 4uv$. The vector $\nabla_H v$ is an eigenvector of Q with eigenvalue $(1 + \kappa)q$, as follows from (5.4). Therefore

$$b = \langle \mathfrak{D}, \varphi \rangle = \int q \{ 6u\psi'(v) + 4(1 + \kappa)uv\psi''(v) \} dg.$$

After angular integration $dx dy = 4\pi du$. Now use $u = t \cos \vartheta$, $z = t \sin \vartheta$ in the right half-plane, and then $v = t^2$. With

$$C_\eta = \int_{-\pi/2}^{\pi/2} \cos^{\eta+1} \vartheta d\vartheta, \quad \gamma = \frac{2-\alpha}{4},$$

integration by parts gives

$$b = 2\pi C_\eta \{6 - 4(1 + \kappa)(\gamma + 1)\} \int_0^\infty v^\gamma \psi'(v) dv = 0.$$

There are no endpoint terms because ψ' is supported away from zero and infinity. The coefficient vanishes because $\alpha = 6\kappa/(1 + \kappa)$. Thus $\mathcal{T}Q = 0$ on the whole group, with no point or line source.

Normalization, powers, and smooth examples. Set $M = \int_{\Omega_0} q dg$, which is finite and positive, and $p = q/M$. Since $Ap = cQ/M$, this is a stationary probability density for the prescribed bounds. On an intrinsic neighborhood $\{u^2 + z^2 < b^2\}$ of the origin, the same polar coordinates give, for every $s > 0$,

$$\int q^s dg = 4\pi \left(\int_{-\pi/2}^{\pi/2} \cos^{\eta s} \vartheta d\vartheta \right) \int_0^b t^{1-\alpha s/2} dt. \quad (5.7)$$

The angular factor is finite and positive. The radial integral is finite precisely when $s < 4/\alpha = s_*$, and diverges logarithmically at equality. Away from the origin, q is bounded on bounded sets. This proves both the claimed subcritical membership and failure at the endpoint.

For completeness, use the group law

$$(x, y, z)(x', y', z') = (x + x', y + y', z + z' + (xy' - yx')/2).$$

Choose a nonnegative $\rho \in C_c^\infty(\mathbb{R}^3)$ of integral one, positive near the identity, and let $\rho_\delta = \delta^{-4} \rho \circ \delta_{1/\delta}$. Define left convolution explicitly by

$$q_\delta(g) = \int \rho_\delta(h) q(h^{-1}g) dh, \quad Q_\delta(g) = \int \rho_\delta(h) Q(h^{-1}g) dh.$$

Left-invariant derivatives in g act on the right factor, so $\mathcal{T}Q_\delta = \rho_\delta * (\mathcal{T}Q) = 0$. Convolution preserves the matrix inequalities and makes both functions smooth, with $q_\delta > 0$. Hence

$$A_\delta = cQ_\delta/q_\delta, \quad M_\delta = \int_{\Omega_0} q_\delta dg, \quad p_\delta = q_\delta/M_\delta$$

are smooth and have all the required coefficient and stationary properties. Local approximate-identity convergence gives $q_\delta \rightarrow q$ in $L^1(\Omega_0)$, so $M_\delta \rightarrow M > 0$. Choose any sequence $\delta_j \downarrow 0$ small enough that $M_{\delta_j} \geq M/2$. For fixed $1 < s < s_*$, Jensen's inequality and Haar invariance bound $\int_{\Omega_0} q_{\delta_j}^s$ by the integral of q^s on one fixed bounded larger neighborhood. This proves (5.2). Pointwise convergence off the axis and Fatou's lemma applied on K_0 give $\liminf_j \int_{K_0} p_{\delta_j}^{s_*} = \infty$, proving (5.1). Finally, $t \log(1 + t) \leq C_s(t + t^s)$ and any fixed subcritical $s > 1$ prove uniform entropy. $\square$

Remark 5.2. For each fixed $L > 1$, a universal local strong L^s theorem for this stationary class on $\mathbb{H}^1$ must have $s < s_*$. The example rules out all $s \geq s_*$ because K_0 has finite volume. It does not identify the largest universally available exponent below s_* , and it does not contradict entropy or a sufficiently small gain above L^1 . These are zero-viscosity examples; smoothness of the approximating coefficients is accompanied by no uniform derivative bounds.